

Machine Learning Based Fault Detection in Electrical Equipment

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Abstract—When I began working on electrical systems, I soon understood the significance of early fault detection in providing safety and stability for the system. Immaterial faults in equipment can have devastating effects such as fires or explosions, threatening both life and property. Early detection, therefore, is not only beneficial—it is a necessity. With the machine learning especially the advanced models like the SVM and KNN we could come up with a scalable method of handling real power systems. The models were highly accurate in predicting and SVM was more than 99 percent accurate. Also confirming to this, we came out with ensemble model by bagging and majority voting which also provided improvement to the detection system. Our method started with both the Particle Swarm Optimization (PSO) and feature selection, thus opening the data with preprocessing followed by tuning of the hyperparameters to advance the performance of the model. The produced system was tested on realistic contemporary datasets and it has uniform reliability under varying contexts. Besides assuring network stability and management of operations for supervisors, customer satisfaction and general optimization of industrial systems—highlighting the vital position of intelligent fault detection in securing and optimizing electrical infrastructures. [1] [2] [8] [9] [12] [15]

I. INTRODUCTION

That is why by definition, a power system is balanced in the normal operating condition. Whenever this is interrupted though e.g. by the insulation breaking down or coming into contact with live running wires a short or fault appears on the line. Such failures could be as a result of other definite causes like weather related occurrences like lightning, high winds, and earthquakes as well as the deterioration of insulation, touching of the wire by trees or the bird hands. Faults can be categorized in to 4 broad types namely Open circuit Fault, Short-Circuit faults, symmetrical and asymmetrical faults. [13] [1]

Moving to Earth With symmetric faults, there is parallel connection of the phases (L L L) or (L L L G). This type of fault is balanced fault. Whenever such a kind of fault occurs, the phase fault currents all become symmetrical i.e. they all have a similar magnitude and are distributed angularly at 120 degrees. It is more serious kind of fault but it rarely occurs infrared thermal imaging have long been used for identifying faults in high-voltage electrical equipment by detecting abnormal heat patterns. [9] [11]

However, these methods often require specialized hardware, are sensitive to environmental conditions, and demand expert interpretation. In contrast, our approach leverages text based machine learning models to diagnose faults using structured data derived from electrical parameters such as current, voltage, power, and resistance.

This method eliminates the need for complex imaging equipment and focuses on data-driven fault detection, which is more scalable

and easier to integrate with modern monitoring systems.. Our system processes numerical and categorical inputs, extracted from real-time sensors and historical logs, allowing for efficient and automated fault diagnosis. This shift from visual to text-based analysis not only simplifies implementation but also enhances the reliability and cost-effectiveness of fault detection in industrial power systems. [18] [9]

In modern-day fast-paced industrial systems, electrical equipment is crucial in the maintenance of stable performance and production efficiency. From my own experiences in monitoring system configurations and dealing with electrical faults, I have witnessed how manual fault detection and conventional method usually find it difficult to cope with the complexity of present control circuit structure. They can miss fault features, causing equipment damage, or even severe safety accidents like fire and explosions. In this case, a judicious strategy based on artificial intelligence tools, deep learning, and machine learning algorithms is used. These techniques make it possible to precisely predict failure modes, such as short circuits, over voltage's, and overloads, with a view to safe operation and reducing economic losses. [12] [15]

II. METHODOLOGY/MODEL

When we initially began this project, we was somewhat perplexed over which dataset to use. The initial data set that we downloaded from GitHub yielded many extra columns, and therefore we made a decision to search for a better one on Kaggle. The data set that we obtained there was not ideal either but was pliable. It contained two extra columns, which we eliminated using the drop method, and we also deal with the missing values by eliminating them to prepare the dataset to be clean and ready for training. The overall strategy that we employed was preprocessing of data, feature engineering, model and assessment. In the beginning of the project, we have cleaned up the data set, addressed the gaps in the data and normalized it in the cases where it was needed, and chosen and built appropriate features which could enhance the model predictive capacity. we then compared and encouraged various machine learning processes, such as logistic regression, decision trees, random forest, and support vector machines, to determine which of them are ideal. After that, we employed the F1 score, recall, precision and accuracy to determine the effectiveness of their functioning.

A. Initial Setup

Importing Libraries: The first cell imports required libraries like NumPy for linear algebra operations, Pandas for data manipulation and analysis, Seaborn and Matplotlib for plots and visualization of data, and different Scikit Learn tools for machine learning purpose. Reading the Data: The dataset is loaded using Pandas, and basic information about the dataset is displayed using df.info().

B. uploading the data set

After importing all the required libraries then we use the `def load data()` command to upload the data set. This function is responsible for reading, displaying, and validating the dataset before any further processing.

C. Data Cleaning and Exploration

Cleaning: The 'id' column is dropped from the dataset, indicating an initial step in data cleaning to remove unnecessary information. **Basic Data Inspection:** The notebook displays the first few rows of the dataset using `df.head(3)`, providing a glimpse into the actual data entries.`df.describe()`. These steps help in understanding the distribution of data and its various properties.

Null Value Check: The notebook includes a check for null values in the dataset using `df.isnull().sum()`. Managing null values Getting the dataset ready for machine learning is a really important step. The next steps would likely include more detailed data visualization, feature engineering, checking correlations, data splitting, and applying machine learning models. **Handling Missing Values.**

D. class distribution.

This box plot shows how the scaled features are distributed in the dataset. On the x-axis, we have the two possible output values, 0 and 1, while the y-axis shows the count or frequency for each of these values. By looking at the plot, we can quickly get an idea of how the data is spread and whether there are any noticeable differences between the two classes, as shown in Figure 1

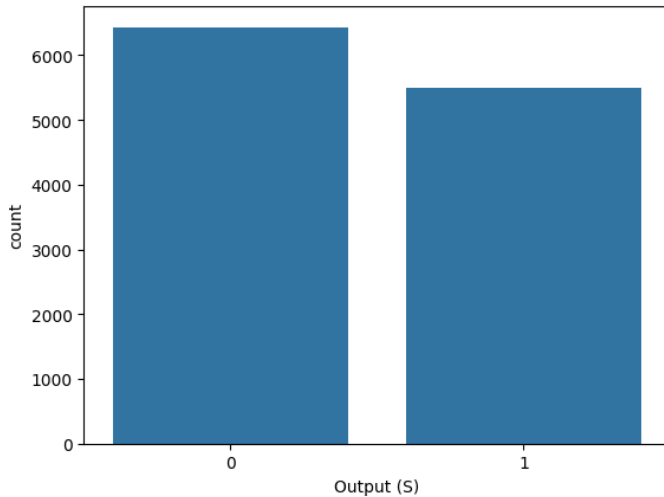


Fig. 1. Class Distribution

E. Correlation Heatmap of Numerical Features

The notebook includes a heatmap to display the correlation between different numerical features. This is a valuable step in understanding how different features relate to each other, which can inform feature selection and engineering, as shown in Figure 2.

F. Feature Engineering

Its next steps likely involve feature engineering, such as encoding categorical variables and scaling numerical values. This is followed by splitting the data, where data is partitioned into training and testing sets, followed by the application of various machine learning models to identify the most precise one.

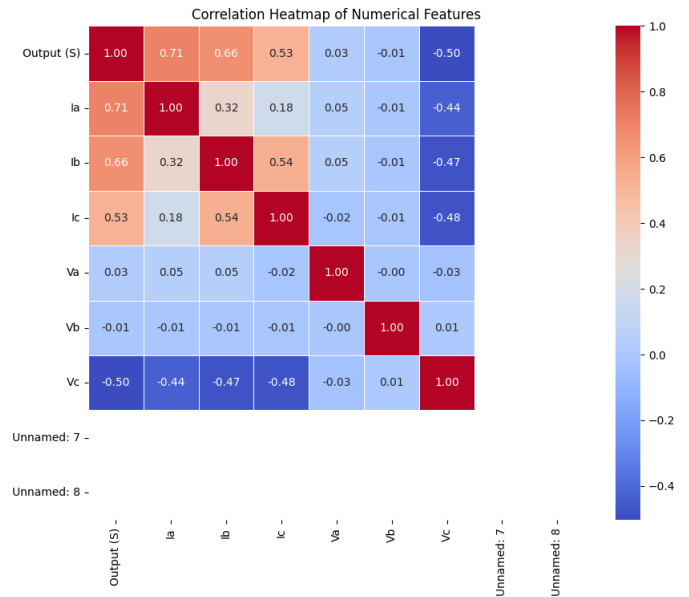


Fig. 2. Correlation Heatmap

G. Box-plot for all numerical features

After this by using some important libraries such as `matplotlib`, `seaborn`, `plotly`, etc. To Visualize data distribution and detect outliers. when i plot this i see there will be many outliers you can see it in Figure ??.

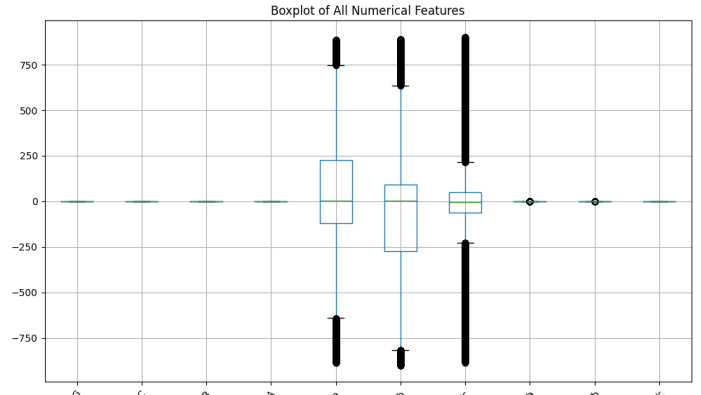


Fig. 3. Box plot with Outliers

H. Handling outliers

The box plot was intended to show the distribution of several numerical variables in a dataset, such as Output (S), Ia, Ib, Ic, Va, Vb, and Vc. However, as we see in the previous figure ??, there are outliers in the dataset shown in the earlier box plot. To handle this, we removed all the outliers from our dataset to improve the accuracy of our model, make the visualization clearer, and speed up the training process.

I. Pie Chart for Fault Types

After handling with outliers with the help of libraries such as `matplotlib`, `seaborn` etc i create a pie chart for fault types it will be helpful at the time of prediction. It will tell the user type of fault

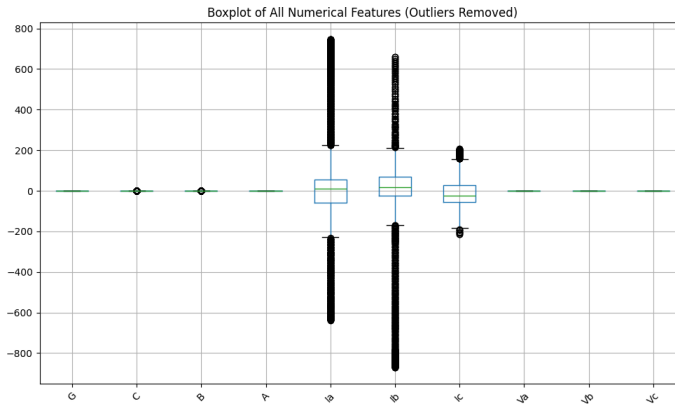


Fig. 4. Box plot without outliers

such as “No Fault”, “Line-to-Ground Fault”, “Line-to-Line Fault”, etc. For this I give different colours to every fault type with encoding data for example for no fault it will be 0000. Also it will help full to Understand class balance for example If one fault type is much more common than others, the dataset is imbalanced. you can see this pie chart in the figure.

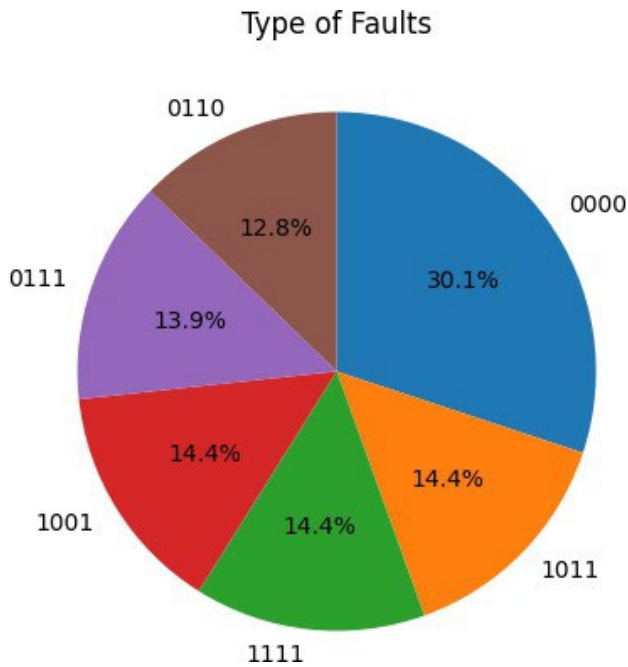


Fig. 5. Pie chart for fault types

J. Scatter Plots for every fault

As in previous section I create a pie chart for each fault after this I used a scatter plot to Visualize the relationship between current (I) and voltage (V) for each phase (A, B, C) during a each type of fault. By using these scatter plot we see how voltage changes with current in a specific phase during the fault. These plots help identify unique patterns for each fault type, making it easier to distinguish between them and understand their impact on the system's

electrical behavior. With the help of some figures we will show the plots for different types of fault. In our model we draw scatter plot for every fault type such as “no fault line-to Ground Fault”, “Line-to-Line Fault”, etc, but we show the scatter plot of any two for more visualization see the figures given below.

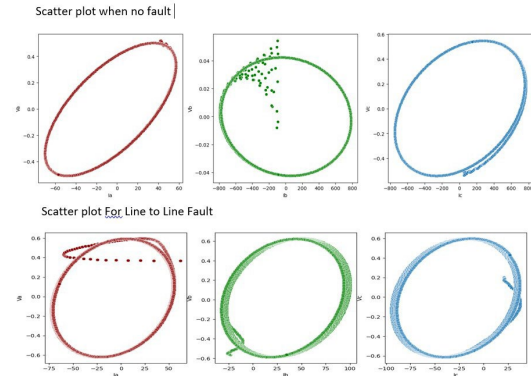


Fig. 6. Scatter plot with no fault and fault

K. Pair Plots of Data

Before moving toward the training and testing of model we decided to create a pair plots of our data set, to see the relationships between features (currents, voltages, etc.). This pair plot is a grid of plots where each column of the dataset is plotted against every other column. It will give us an early visual check before training our model. you can see it and visualize it in the figure given below.

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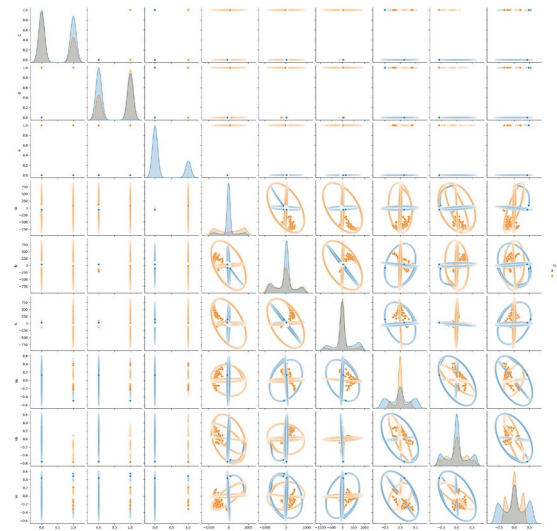


Fig. 7. classification report

L. Train-Test Split

To evaluate the performance of our machine learning model effectively, we divided the dataset into two parts: 80 and 20. The model has enough data to learn from while still preserving a portion of unseen data for performance evaluation. Training the model on the larger part allows it to identify patterns, relationships, and key features within the data.

M. Model Training

Once the classifier is initialized, using the training portion of the dataset we move toward the training part of our model when we start we train over data set with different model such as SVM,KNN ,Decision Tree Model, Random Forest Classifier etc. But we have many issue with our prediction also have issue with confusion matrix due to this we are facing issue with classification report of our model we train our data set with different model but we have issue with classification report that is some time we achieve good accuracy but we have low precision recall f1 score etc.

N. Model Evaluation

After data splitting, several machine learning models are trained on the training set. These models are compared on the basis of how accurately they predict on the test set. This assessment phase is the most significant in any machine learning pipeline since it indicates the actual-world Performance of the model. The assessment, which can be carried out on the basis of such indicators as accuracy, precision, recall, and F1-score, allows us to understand the advantages and some flaws of each model-and, apparently, the first time, to choose the most effective one to help us to accurately identify faults in electrical equipment.

1) *Confusion matrix*: In the notebook, you will also see a confusion matrix that helps us visualize how well the model is doing. This graphical tool compares the true labels to the predicted labels, providing us with a clear view of true positives, true negatives, false positives, and false negatives. It gives us insights beyond accuracy, highlighting where the model would be incorrectly classifying some faults. From this matrix, we are able to identify certain areas where the model could be lacking some errors, particularly false negatives—instances in which the model does not identify a real fault. Identifying these problems enables us to refine the system,making it more robust and trusty for fault detection in electrical machines..

2) *Performance Metrics*: Apart from the confusion matrix, the notebook also computes other important performance metrics like classification accuracy and error rate. These are useful in assessing how well the model is doing in general and where it can be improved.

III. RESULTS

A. Precision

Precision indicates to us the proportion of positive predictions made by the model that turn out to be correct. It's the ratio of true positives to the number of times the model predicted a positive result, which consists of both true positives

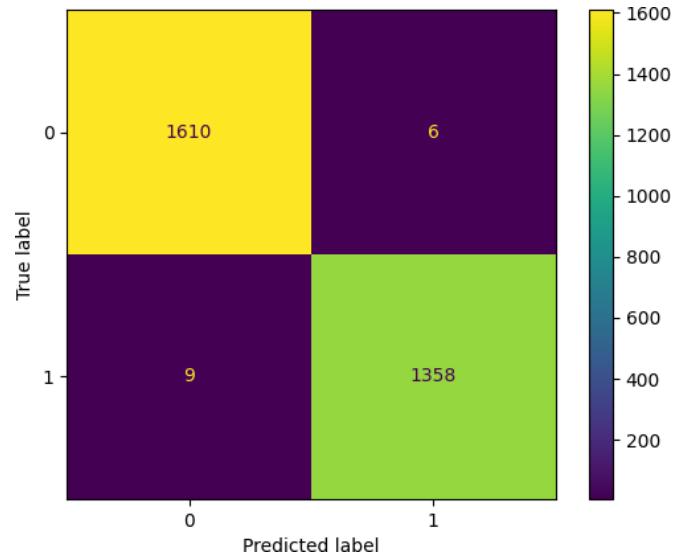


Fig. 8. confusion matrix

	precision	recall	f1-score	support
0	0.99	1.00	0.99	1616
1	1.00	0.99	0.99	1367
accuracy			0.99	2983
macro avg	0.99	0.99	0.99	2983
weighted avg	0.99	0.99	0.99	2983

Fig. 9. classification report

and false positives. This appears us with how trustworthy the model's positive predictions are. Here, for class 0 (which might be "no fault"), the precision is roughly 0.99, i.e., nearly all of the times when the model predicted class 0, it was correct. For class 1 (which may be a fault), the precision is even more—roughly 1.0—indicating that almost every fault prediction by the model was correct.

B. Recall

Recall, alternatively referred to as sensitivity or the actual positive rate, indicates the quality of the model in detecting all the true positive cases. It rates the ratio of true positives to all true positive instances, including the false negatives that were not caught by the model. This measures how well the model performs in capturing positive cases. In this case, class 0's recall is approximately 1.0, which means the model was able to identify nearly every actual case of class 0 correctly. For class 1, the recall is approximately 0.99 and indicates the model is highly proficient at discovering nearly all the actual positive cases in class 1 as well.

C. F1-score

The F1-score takes both precision and recall and puts them into one number, providing an overall assessment of how well the model does. For class 0, the F1-score is approximately

0.99, and for class 1, it's also about 0.99. This indicates the model is performing well both in accurately classifying positives and also not missing them.

D. Accuracy

Accuracy reflects how often the model predicts correctly when considering all the data, including all the classes in the data. It provides a basic overall impression of the model's performance by indicating the proportion of total predictions that were correct. The model in this instance has an accuracy of approximately 0.99, which means the model gets the class correct nearly always. This high level of accuracy means that the model is consistent and reliable in its predictions in varying situations and is thus a good prospect for fault detection in electrical machinery in real life.

IV. FUTURE IMPROVEMENTS

In the future, the project can be enhanced by incorporating real-time monitoring using IoT so that there can be ongoing data collection and analysis. Establishing an API will enable the system to be accessed remotely in a safe and efficient manner. The dataset may also be increased with additional real-world measurements so that the model will be more precise and reliable. On top of that, investigating more advanced deep learning architectures may unlock even higher performance.

V. CONCLUSION

This detailed analysis of the notebook shows a well-considered well-rounded and systematic approach to fault prediction in electrical devices through machine learning. The project covers all the relevant steps, from data preparation to analyzing it thoroughly, applying feature engineering methods, training numerous models, and testing in detail how well they work.

Having accurate visualizations at every step gives rich insights into the data and model behavior. In addition, the choice of multiple evaluation metrics ensures that the model's performance is calculated from different aspects, resulting in a robust and stable fault detection system.

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